RetCam Imaging for Retinopathy of Prematurity Screening

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Purpose: Indirect ophthalmoscopy is the gold standard for retinopathy of prematurity (ROP) screening. Screening for ROP with digital imaging has been proposed as a possible alternative. Our goal was to evaluate the longitudinal clinical outcomes of employing digital imaging to detect high-risk ROP. ***Methods:*** Serial RetCam imaging and indirect ophthalmoscopy were performed on 43 premature infants. A masked reader evaluated the images and made management recommendations that were compared with indirect ophthalmoscopy results. Successful screening was determined by correctly identifying progression to prethreshold or threshold disease with referral for indirect ophthalmoscopy. Unsuccessful screening was determined by failure to identify prethreshold or threshold disease, inaccurately detecting prethreshold or threshold disease, or inability to evaluate for ROP. ***Results:*** No cases of prethreshold or threshold disease were missed by the reader. The reader overestimated prethreshold or threshold disease in 5% of cases. Initial screening in 21% of cases could not be evaluated for ROP secondary to poor image quality. Digital photography had a sensitivity of 100% and specificity of 97.5% in detecting prethreshold and threshold ROP. Positive-predictive value of digital photography was 67% and negative-predictive value was 100%. ***Conclusions:*** Screening and management of ROP using RetCam imaging did not fail to detect prethreshold or threshold disease when images could be obtained. Ophthalmologic examinations were needed in 20% of cases that did not reach threshold or prethreshold disease because of poor image quality or overestimation of ROP. RetCam screening may safely reduce the overall number of indirect ophthalmologic examinations required. (J AAPOS 2006;10:107-111)

Retinopathy of prematurity (ROP) is a potentially blinding eye disease and a leading cause of vision loss in children. Each year ROP affects an estimated 14,000 to 16,000 premature, low birth weight infants in the United States.[1] Approximately 1500 of these infants will develop threshold ROP requiring treatment and 400 to 600 infants will become legally blind despite treatment.[1]

Early detection and treatment of threshold and high-risk prethreshold ROP have been shown to significantly decrease the incidence of severe vision loss and reduce unfavorable outcomes in premature infants.[2,3] Current ROP screening guidelines in a joint statement by the American Academy of Ophthalmology, the American Association for Pediatric Ophthalmology and Strabismus, and the American Academy of Ophthalmology state that infants with a birth weight of less than 1500 g or a gestational age of less than 28 weeks, as well as selected infants between 1500 and 2000 g with an unstable clinical course who are believed to be at high risk by their attending pediatrician or neonatologist, should have at least two fundus examinations performed after pupillary dilation using binocular indirect ophthalmoscopy to detect ROP.[4] The examination for ROP should be performed by an ophthalmologist with sufficient experience and knowledge in the examination of preterm infants for ROP using binocular indirect ophthalmoscopy.[4]

In the Multicenter Trial of Cryotherapy for Retinopathy of Prematurity (CRYO-ROP) study, 6% of infants weighing less than 1251 g developed threshold ROP.[5] Recently, the Early Treatment for Retinopathy of Prematurity (ETROP) study has suggested revised guidelines for treatment of high-risk prethreshold ROP that would increase treatment to approximately 8% of these infants.[3]

Indirect ophthalmoscopy examinations are labor-intensive for the ophthalmologist, stressful for the infants, and often unavailable in remote or underserved areas. Given the low treatment yield of all screened infants, alternatives to indirect ophthalmoscopy for ROP screening have been sought.

As telemedicine is becoming more widespread, digital cameras that reliably reproduce retinal images of infants will have an impact on the screening and management of ROP. Previous studies have focused on whether digital

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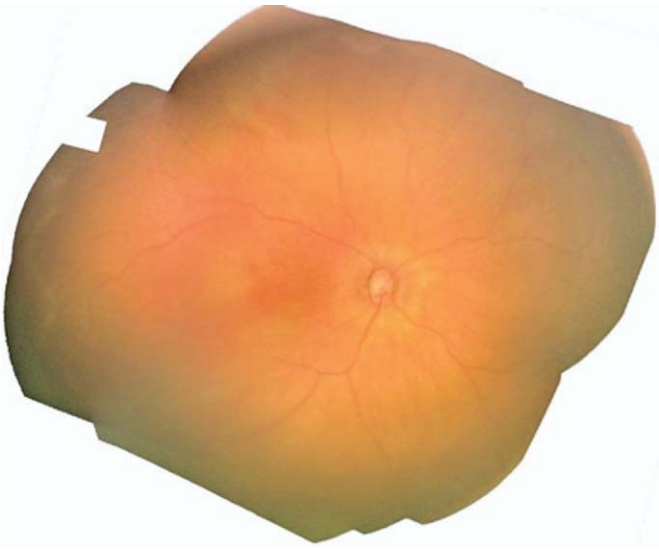


FIG 1. Composite RetCam image of stage 1 ROP.

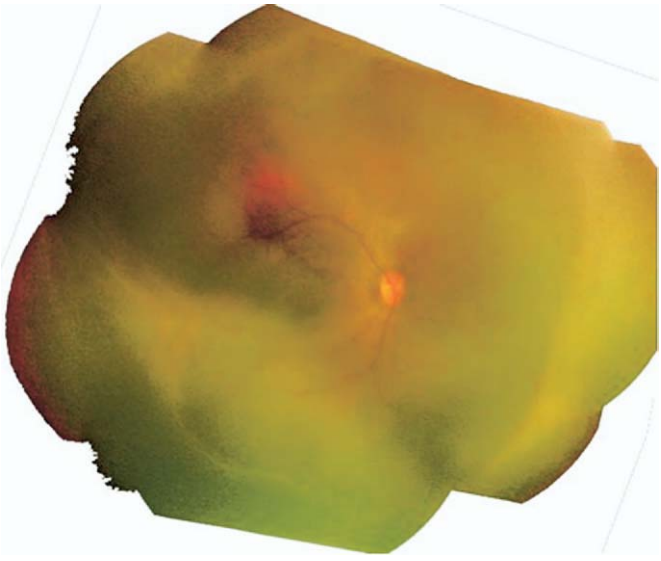


FIG 2. Composite RetCam image of stage 3 ROP.

photography can detect all stages of ROP, the optimal timing of digital photography, or the evaluation of single digital images.[6-9] Our goal was to evaluate the longitudinal clinical outcomes of employing digital imaging to screen for severe ROP requiring treatment.

METHODS

Patients

Premature infants undergoing routine ROP screening examinations at Children's Hospital Boston and Brigham and Women's Hospital Neonatal Intensive Care Units (NICUs) were evaluated from August 2003 to January 2004. Screening guidelines for ROP at these two institutions included infants with gestational age (GA) less than 32 weeks, birth weight (BW) less than 1500 g, or older and heavier babies with an unstable clinical course who were believed to be at high risk for ROP by their attending neonatologist. Infants were excluded from this study if they had major ocular anomalies or media opacities.

Examination Schedule

The first ROP examination was performed according to the following schedule—6 weeks of age for babies born younger than 26 weeks GA, 5 weeks of age for babies born between 27 and 28 weeks GA, 4 weeks of age for babies born between 29 to 30 weeks GA, and 3 weeks of age for babies born 31 weeks GA and older. Thus, the first screening was performed between 30 and 34 weeks GA. Subsequent examinations were scheduled biweekly if no ROP was present and weekly if ROP was present. Screening was continued until the infant was discharged or transferred from the unit, the infant required treatment, or the retinal vasculature was mature.

Examination Techniques

The infant's pupils were dilated with one drop of combined 0.2% cyclopentolate and 0.1% phenylephrine (Cyclomydril, Alcon, Fort Worth, TX) and one drop of 1% cyclopentolate (Cyclogyl, Alcon) at least 30 minutes prior to examination. Topical proparacaine was applied and an eyelid speculum was inserted. Digital images were taken with the RetCam Digital Retinal Camera (Massie Research Laboratories Inc., Pleasanton, CA) using the 130° ROP lens immediately prior to the indirect examination. The goal of the RetCam examination was to obtain an evaluable image of the posterior pole and each of the four quadrants. No more than 2 minutes, and usually <1 minute, were spent imaging each eye. The RetCam examinations were performed by a study ophthalmologist (C.W. or D.K.V.) or a skilled technician. A series of 1 to 10 photographs were taken. The images were stored on the hard-drive of the RetCam machine.

Indirect ophthalmoscopy with a 28-D lens and scleral depression was then performed by the study ophthalmologist (C.W. or D.K.V.) who was present during the RetCam imaging. The presence or absence of ROP and plus disease were recorded. If ROP was present, the zone, stage, and extent of ROP were also recorded.

Reading of Digitized Images

RetCam images were transferred via a CD to another computer. Composite images, when possible, were created using the EyeTool Kit (EDC Lamy, Carvin, France) (Figures 1 and 2). Identifying patient data were removed from these composite images and the images were saved on a CD.

Images were viewed from the CD by a single reader (R.A.P.) skilled in ROP examinations. The reader was given the patient's GA, BW, race, sex, birth multiplicity,

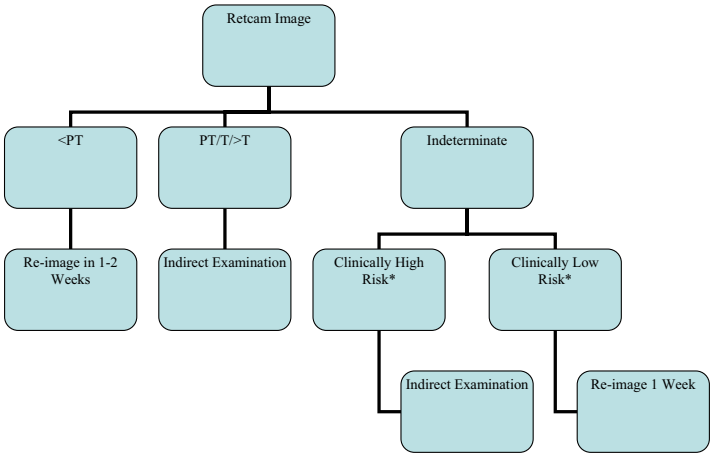


FIG 3. Management recommendations for RetCam ROP screening.

and current postconception GA (CGA). The reader evaluated for ROP and plus disease.

ROP was graded as indeterminable, less than prethreshold, prethreshold, threshold, or greater than threshold. Prethreshold disease was defined as any zone 1 ROP, zone 2 stage 2 ROP with plus disease, zone 2 stage 3 ROP without plus, or zone 2 stage 3 ROP with plus disease but less than the requisite clock-hours to qualify as threshold disease. Threshold disease was defined as five contiguous or eight cumulative clock-hours of stage 3 ROP with plus disease in zones 1 or 2. Greater than threshold disease was defined as stage 4 or 5 ROP. The masked reader then made management recommendations (Figure 3)-re-imaging at the recommended interval for less than prethreshold disease, or referral for indirect examination when the infant reached prethreshold or threshold disease. If the reader could not evaluate the image, recommendations were made either to repeat the RetCam examination at a specific interval or to request an indirect examination at a specific interval depending on the clinician's assessment of the patient's clinical ROP risk. Recommendations were compared with indirect ophthalmoscopy results and clinical outcomes. We did not compare exact examination findings such as number of clock-hours or stage of disease. Our goal was to identify severe ROP requiring treatment, not whether RetCam and indirect ophthalmoscopy were able to exactly correlate with each other.

Successful screening was defined as correctly identifying progression to prethreshold or threshold disease with referral for indirect ophthalmoscopy or correctly identifying less than prethreshold disease with no referral for indirect ophthalmoscopy. Unsuccessful screening was defined as failure to identify prethreshold or threshold disease, inaccurately detecting prethreshold or threshold disease with non-disease-warranted referral for indirect ophthalmoscopy, or inability to evaluate for ROP.

RESULTS

ROP screening with RetCam imaging and indirect ophthalmoscopy was performed on 86 eyes of 43 infants. GA ranged from 23 to 33 weeks with a mean GA of 27.3 weeks. BW ranged from 460 to 2290 g with a mean weight of 1024 g. In this cohort, 42% of the infants developed ROP and 5% of infants required treatment.

No RetCam imaging had to be aborted secondary to patient stress. Initial images in 21% of cases could not be evaluated for ROP secondary to poor image quality (Figure 4). However, in 78% of these cases subsequent digital imaging was adequate for ROP evaluation.

No cases of prethreshold or threshold disease were missed by the reader. In 5% of the cases, the reader overestimated prethreshold or threshold disease with incorrect referral for an indirect examination. No cases of plus disease were missed by the reader.

Digital photography had a sensitivity of 100% and specificity of 97.5% in detecting prethreshold and threshold ROP. Positive-predictive value of digital imaging to detect prethreshold or threshold ROP was 67% and the negative-predictive value was 100%.

DISCUSSION

With the advent of telemedicine and the decreasing number of ophthalmologists skilled in or available for examining infants for ROP, screening and management of ROP by digital imaging has been proposed. Previous studies have evaluated the sensitivity and specificity of detecting all stages of ROP and have shown that ROP in peripheral zone 2 or 3 was often missed in RetCam

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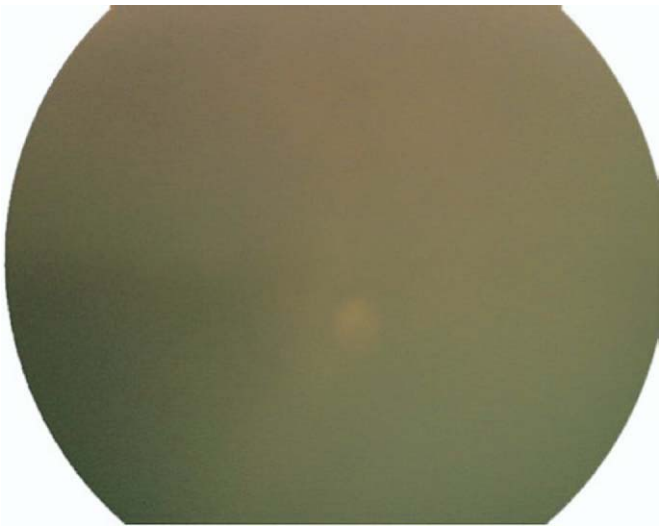


FIG 4. RetCam image from a darkly pigmented infant.

evaluations.[6-8] Schwartz et al[9] compared telemedical evaluation and management with traditional on-site evaluations of 10 patients with ROP. Plus disease was accurately identified in 95% (18/19) of eyes and the presence of prethreshold, threshold, and stage 4 or 5 ROP was correctly identified in 89% (17/19) of eyes. Additional studies have shown a good correlation between RetCam interpretations by neonatologists and ophthalmologists[10] and that RetCam imaging is useful for documentation, follow-up, and teaching in addition to conventional ophthalmoscopy.[11]

For this study, we defined successful RetCam screening as the ability to detect ROP that required or was likely to require treatment (prethreshold or threshold ROP). We did not spend additional time or induce added stress on the infants by attempting to document very peripheral ROP, which has already been shown to be difficult or impossible to image. By reliably detecting cases of high-risk ROP, a significant number of NICU indirect ophthalmoscopy screenings that do not warrant treatment could be safely reduced. In our study, no cases of prethreshold or threshold disease were missed. Conversely the reader tended to overestimate the severity of the disease. The sensitivity and specificity of the RetCam images in detecting severe ROP in our study were similar to those found by Ells et al.[12] Their criteria for severe disease included any ROP in zone 1, the presence of plus disease, or the presence of any stage 3 ROP. In their study, digital imaging had a sensitivity of 100% and a specificity of 92% in detecting high-risk ROP.

Given that ROP that requires treatment necessitates either zone 1 disease (which has been shown to be easily imaged) or plus disease (which is diagnosed in the posterior pole), it is not surprising that no cases of prethreshold or threshold disease were missed. Our RetCam examinations were limited in that 21% of our initial screenings could not be evaluated for ROP secondary to poor image quality, necessitating referral for indirect ophthalmoscopy. Heavily pigmented infants and infants <32 weeks GA were difficult to image secondary to hazy media. Additionally, small palpebral fissure size in some infants prevented adequate contact between the RetCam lens and the corneal surface. Subsequent RetCam imaging with readable images was possible in 78% (7/9) of these infants.

Some infants had only one RetCam screening due to being discharged from the NICU prior to the second follow-up examination. We included these patients because the goal of our study was to evaluate whether RetCam screening could limit the number of NICU examinations using indirect ophthalmoscopy. Once the infants are discharged from the hospital, they are better able to travel to an ophthalmology appointment and are more medically stable to tolerate examinations.

There are advantages and disadvantages of both types of examinations. Advantages of indirect ophthalmoscopy include more complete documentation of ROP and usually better visualization of the fundus by the ophthalmologist; disadvantages include availability and time constraints of the ophthalmologist. RetCam screening may be advantageous in that it may be performed by a technician or nurse with possibly more flexibility in scheduling, although disadvantages include limitations of image quality and complete detection of ROP. Another disadvantage of RetCam screening includes the initial cost of the equipment, although over time this may be balanced by decreased costs of ophthalmology referrals. A cost analysis of RetCam imaging should be performed, but was not a part of this study.

This study was conducted prior to the publication of the new treatment guidelines of the ETROP Study. Even so, all characteristics of eyes for which early treatment might be considered (zone 1 ROP and zone 2 ROP with plus disease) would have been detected using our screening techniques.

We believe that RetCam imaging is a useful screening tool to detect treatable ROP and may safely reduce the overall number of indirect ophthalmoscopy examinations required. RetCam imaging was able to accurately detect cases of high-risk ROP (prethreshold and threshold disease) requiring indirect ophthalmoscopy, thus allowing low-risk ROP to be safely screened through RetCam examinations. However, the RetCam image reader must take into careful consideration the infant's gestational age and RetCam findings in deciding follow-up recommendations. We are not advocating that RetCam imaging completely replace indirect ophthalmoscopy. To fully assess whether an infant's retina is mature and normal, an indirect examination should be performed at least once, even upon discharge from the NICU.

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An Eye on the Arts – The Arts on the Eye

7 Opening eyes deprived of light,
Rescuing prisoners from confinement,
From the dungeon those who sit in darkness.
16 I will lead the blind
By a road they did not know,
And I will make them walk
By paths they never knew.
I will turn darkness before them to light,
Rough places into level ground.
These are the promises—
I will keep them without fail.
18 Listen, you who are deaf;
You blind ones, look up and see!
19 Who is so blind as My servant,
So deaf as the messenger I send?
Who is so blind as the chosen one,
So blind as the servant of the Lord?
20 Seeing many things, he gives no heed;
With ears open, he hears nothing.
21 The Lord desires His [servant's] vindication,
That he may magnify and glorify [His] Teaching.

—from *Isaiah*